GENERAL SPECIFICATION G6.2

Installation Requirements

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G6.2 INSTALLATION REQUIREMENTS

G6.2.1 ICA Cabling

(a) Introduction

(i) The Contractor shall provide an ICA cabling system comprising signal, control, UPS power and non-UPS power to all ICA equipment.

(ii) All cables shall be supported using cable trays/ladders.

(iii) All cables shall be delivered on robust cable drums with cable ends treated to form an effective seal. When a cable is cut from a drum the cable end and the end left on the drum shall be immediately sealed in a manner appropriate considering the expected interval until the next usage of the drum, with regard to the prevention of the ingress of moisture.

(iv) Suitable segregation between various types of cabling shall be maintained according to the Electrical section of the Specification to ensure there is no risk of electromagnetic interference. Unless otherwise specified, the minimum space between cables shall be as follows:

ELV control cables to high voltage cables	:	300 mm
ELV control cables to low voltage power cables	:	150 mm
Instrumentation, fieldbus, telephone cables to high voltage cables	:	300 mm
Instrumentation, fieldbus, telephone cables to low voltage cables	:	150 mm
ELV control cables to LV control cables (which appropriate insulation)	:	nil

(v) Telephone cables run separately in fully enclosed metal trunking or metal conduit, within buildings, may have a reduced separation from parallel runs of LV power cables of 50 mm.

(vi) Whenever instrumentation and power cables cross path while laying, they shall cross at 90 degrees so that the electro-magnetic interference due to crossing becomes zero.

(vii) Where cables are installed in close proximity to or cross those of external authorities such as telecommunication, electricity supply and railway, spacing shall comply with the regulations of those authorities.

(viii) The Contractor shall ensure that all cables are of an adequate cross-sectional area for the application (subject to the minimum size specified).

(ix) The Contractor shall provide a comprehensive system for the support, accommodation and protection of all cabling provided under the Contract. Cabling under a raised floor or over a suspended ceiling shall be supported by cable ladder rack or cable tray. The cabling design shall be based on the following requirements:

1) Cabling to control desks shall enter through the base of the desk; and

2) For panels located indoors, cabling shall generally enter panels and enclosures through the bottom.

(x) These ICA cable specifications shall be read in conjunction with the Electrical section of the Specification.

(xi) Refer to Electrical Specifications for requirements of floor openings, box-outs, plinths, cast-in ducts, and multi-cable transits. ICA cable installation requirements including but not limited to glanding, support and labelling, shall be as per the Electrical section of the Specification.

(xii) Cable glands shall be provided to allow cable entry to the instrument enclosures without degrading the required degree of protection.

(xiii) Where required by system manufacturer's recommendations, network and field bus cable glands shall be earthing type, to provide earthing of the cable screen at the entry of the control cubicles.

(xiv) Use of a "common" (e.g. common +24 V dc or common 0 V dc) within multi-core or multi-pair cables is not preferred. The Contractor shall highlight their intended use of commons in their design submissions and the S.O. shall retain the right to insist to dedicated wires in any instance. Where the use of commons is approved the contractor shall ensure that the additional loading on the common does not result in excessive voltage drop, nor lead to loss of both duty and standby signals or measurements in the event of wire failure.

(xv) Signal levels shall not be mixed within multi-pair or multi-core cables though inputs and outputs (of same level) may be combined.

(xvi) For intrinsically safe cables, the Contractor shall ensure that the cable inductance does not exceed the limits for intrinsically safe circuits.

(xvii) Cables in intrinsically safe circuits shall preferably not be run in the same tray as cables in non-intrinsically safe circuits. If run in the same tray, an earthed metallic separator shall be provided.

(xviii) Where cables are run through conduit, the entry and exit shall be smooth and free from burrs.

(xix) Cables must be pulled into conduit in a way that ensures there is no damage to the cable.

(xx) Signal isolators shall be provided for all analogue signals having a common-mode voltage differential (to other signals in the same panel) exceeding 10 volts.

(xxi) Cable junctions shall be made only at terminals inside instruments or approved junction boxes. In any circuit there shall be no more than one junction point.

(xxii) Instrument cable road crossings shall be in form of duct banks in which conduits are covered in concrete designed to take load of the vehicular movement on the road.

(xxiii) ICA cabling installed at the chemical areas/storages, shall have sheath material suitable for the chemical environment.

(b) Telecommunication Cables

(i) Telecommunication cables shall comply with IEC 60708.

(ii) The conductor shall be of high conductivity, plain, multi-stranded annealed tinned copper.

(iii) The insulation shall be of solid polyethylene compound. The insulation colours shall be easily identifiable.

(iv) There shall be a minimum of 2 pairs in any cable.

(v) Cables shall be individually screened by laminated aluminium / PETP tape. Pairs shall be disposed with different layers to keep cross talk between pairs to a minimum. The interlayer lapping tape shall be of polyethylene terephthalate.

(vi) Where specified, armouring shall be of a single-layer galvanized steel wire laid helically over the sheathed cable. The armour shall be protected against corrosion by an extruded over sheath of black PVC.

(vii) The over sheath shall be extruded as a continuous, seamless, close fitting tube of PVC. The sheath shall be black colour, termite proof and where specified shall have flame retardant characteristic.

(c) Signal Cables

(i) Single-pair analogue signal cables shall be steel wire armoured, shielded twisted pair cables with a minimum 1.5 mm^2 conductors, tinned, SWA with a minimum insulation grade of 500 V. Conductors shall be multi-stranded 98% annealed tinned copper and the coverage of the screen shall be at least 80%. Core colours shall be white and black. Contractor shall calculate the voltage drop considering the distance between the field instrument and IO card. Voltage drop shall not exceed 10% unless the supplied equipment is designed to operate at a corresponding lower voltage.

(ii) Cables for multiple pair analogue signals shall be steel wire armoured multi-pair cables. These shall be as for single pair cables but shall have tinned conductors of a nominal 0.5 mm^2 with both element (individual pair) and overall screens.. Multi-pair cables shall have 20% spare pairs or two spare pairs whichever is greater. Contractor shall calculate the voltage drop considering the distance between the field instrument and IO card. Voltage drop shall not exceed 10% unless the supplied equipment is designed to operate at a corresponding lower voltage.

(d) Control Cables

(i) Control cables for digital signals shall be a minimum of 1.5 mm^2 tinned PVC/SWA/PVC with a minimum insulation grade of 600/1000 V with individually numbered (continuous black printing on white insulation) cores. Control cables shall have 20% or two spare cores whichever is greater except for final cables to field devices having only one signal for which a minimum of one spare core (i.e. 3 cores total) shall be acceptable. Control cables shall have either four, seven, twelve, nineteen, twenty-seven or thirty-seven cores. Voltage drop shall not exceed 10% unless the supplied equipment is designed to operate at a corresponding lower voltage.

(e) Fibre Optic Cable

(i) Single-mode cables in shall be 900 μm tight-buffered. They shall have an inner buffer of UV-cured soft acrylate, and an outer buffer of low smoke halogen free (LSHF) material. The fibres shall be graded index 8.3/125 μm fibre having an attenuation of not greater than 0.50 dB/km @ 1300/ 1550 nm. There shall be 100% spare fibres in the form of an entirely spare cable. All fibres (including spares) shall be terminated at both ends to allow future use.

(ii) Multi-mode cables shall be 900 μm tight-buffered. They shall have an inner buffer of UV-cured soft acrylate, and an outer buffer of low smoke halogen free (LSHF) material. The fibres shall be graded index OM3 50/125 μm fibre having an attenuation of not greater than 3.0 dB/km @ 850 nm or 1.0 dB/km @ 1300 nm. There shall be 100% spare fibres in the form of an entirely spare cable. All fibres (including spare cores) shall be terminated at both ends to allow future use.

(iii) Fibre optic cables shall be FRP armoured for rodent protection and shall have a Nylon (Polyamide) jacket for termite resistance.

(iv) Fibre optic cables shall be terminated inside suitable FOBOTs and patch panels at each location. Patch panels shall use “LC” type connectors with ceramic ferules.

(v) Any fibre optic splices shall use fusion splicing.

(f) Copper Communications Cable

(i) Foil Twisted Pair (FTP) shall be used for data cable which shall comply with the following:

1) Category 6A FTP, UL listed, and third party verified to comply with TIA/EIA 568 requirements;

2) Suitable for high speed network applications including gigabit Ethernet;

3) Flammability plenum requirements of NFPA 70 and NFPA 262; and

4) Cable shall withstand a bend radius of 1 inch minimum at a temperature of minus 20 degrees C maximum without jacket or insulation cracking.

(g) Copper Communications Cable

(i) Unshielded Twisted Pair (UTP) shall comply with the following:

1) Category 5E UTP, UL listed, and third party verified to comply with TIA/EIA 568 requirements;

2) Suitable for power over Ethernet applications;

3) Flammability plenum requirements of NFPA 70 and NFPA 262; and

4) Cable shall withstand a bend radius of 1 inch minimum at a temperature of minus 20 degrees C maximum without jacket or insulation cracking.

(h) Modbus (RS-485) Cabling

(i) Meet or exceed electrical characteristics of EIA-485 Standard;

(ii) Meets recommendations of MODBUS hardware manufacturer;

(iii) Outer Jacket: PVC, low smoke halogen free insulation; and

(iv) Conductors:

1) One twisted pair 20 AWG, seven-strand tinned copper;

2) Insulation: Polyethylene; and

3) Core Colours: Blue and white.

(i) Underground ICA Cables

(i) Where ICA cables are installed underground, they shall be run in uPVC Class B ducts of suitable diameter.

(ii) Unless approved otherwise by the S.O., 100% spare ducts shall be provided complete with draw-pits and draw wires as well as corresponding building entries; and

(iii) Refer also to General Specification G5 Electrical for other requirements pertaining to underground cables.

G6.2.2 Instrument Piping, Tubing and Fittings

(a) Materials

(i) Instrument impulse lines shall be in accordance with the manufacturer's recommendations for the specific applications. All connections from the process to instruments other than water quality meters shall be made with 316 seamless stainless-steel tubing of 6 mm O/D, minimum wall thickness 1.5 mm.

(ii) The main instrument air distribution network shall be as specified in mechanical sections of the Specification. Sub-headers and tubing shall be as specified in the following clauses.

(iii) Connection points to the main air distribution line shall be done using the highest available point of the air distribution line, to minimize any condensation or moisture contamination to the instrument/actuator.

(iv) Instrument air supply pipework greater than 25 mm diameter shall be hot-dipped galvanised carbon steel. Instrument air pipework below 25 mm diameter shall be 316 stainless steel tubing.

(v) For groups of 2-5 actuators, the instrument air sub-header shall be 15 mm NB minimum and for groups of 6-20 actuators, 25 mm NB minimum.

(vi) For each actuator, a 12-mm block valve, filter and regulator shall be provided irrespective of the size of the supply header.

(vii) Fluid sample pipework and fittings for analysers shall be sized and selected to suit service conditions generally in the range 12-25 mm NB and shall be in 316 stainless steel. In the case of dissolved oxygen meters, pipework and fittings shall be in 316 stainless steel only.

(viii) Connections of pressure gauges, transmitters and switches to the process shall incorporate pulsation dampers where appropriate.

(ix) All instruments and analysers shall be provided with a three-way or five-way (as appropriate) valve manifold for isolation, drainage and connection to calibration equipment.

(x) Pipework to differential pressure operated devices shall include a five-valve manifold near the device. Pipework to steam, oil and high-pressure lines shall have two isolating valves near the instrument. Tapping points for instruments connected by impulse pipework shall have additional isolating cocks. Level meters used with an air-reaction system shall be installed at the highest point of the piping. Variable area meters shall have isolating valves upstream and downstream with, in the case of liquid measurement, a drain valve between the upstream valve and the meter.

(b) Instrument Piping and Tubing - Routing and Support

(i) Piping shall be neatly run and shall be clipped to walls, ceilings or other building structures or shall be supported on cable tray of suitable material. Piping routes shall be chosen to avoid obstructing traffic or personnel movement through the plant and to avoid interference with accessibility for the removal of plant. Piping shall be routed away from hot environments, places of potential fire risk or mechanical abuse. Piping routes shall also avoid areas where liquid spillage or vapour or gas leakage could occur. Piping and supports shall avoid vibrating structures or equipment. Piping shall slope continuously upwards or downwards with a minimum slope of 1 in 10 and in accordance with any other recommendation of the equipment manufacturer and sharp bends shall not be used. Drain and vent facilities shall be provided at the lowest and highest points respectively according to service. Drains and vents shall be plugged.

(ii) Pipework shall be arranged to minimise the transmission of vibration. Pipework to steam lines or vessels shall contain a siphon, loop or condensate chamber and shall be lagged. Each instrument and accessory shall be supported independently of its pipework.

(iii) All pipework for fluids which could contain solid particles shall contain joints and bends of the "cross" type for rodding through and shall contain valves and connections for blowing down. The length of any line under vacuum shall be as short as possible. Gas analyser sample venting lines shall be sloped downwards away from the instrument or other precautions taken to prevent condensate flowing back to the cell. In the case of toxic or flammable gases, venting shall take place in an approved manner.

(iv) Impulse lines for flow measurement shall be run in accordance with BS 1042. Tubing shall be clipped to mild steel, galvanised tray and shall not be pierced, stretched or twisted. The minimum bending radius shall be 100 mm. Any excess length shall be neatly coiled at the measuring instrument and clipped in coils of not less than 200 mm diameter.

(c) Seals

(i) Process fluids which solidify or are highly viscous at the lowest ambient temperature shall be prevented from entering impulse lines and instruments using seal pots, which shall be filled with a suitable inert, non-toxic liquid which shall not evaporate under process conditions, solidify, mix or react with the process fluids. If large displacements of fluid are possible, seal pots shall be provided close to the process connections. Seal pots shall have isolating, filling and venting valves.

(d) Instrumentation Pipe Blowing

(i) Before connecting any instrument to its associated impulse or air supply pipework, all isolating valves shall be opened and the pipework shall be blown through to clear any matter. Each line shall be flow-tested for continuity, followed by a pressure test at 2 bar, or as otherwise specified using clean dry air. Instruments and other ancillary equipment shall be connected immediately after the successful completion of the pressure test, if possible, and all open tube ends shall be sealed against the ingress of moisture.

G6.2.3 Lightning and Surge Protection for ICA Equipment

(a) The protection systems shall be such that the protective level shall not interfere with normal operation, but shall be lower than the instrument surge withstand level.

(b) Each surge protection device shall be connected to a suitable earth.

(c) Surge protection devices shall be provided for all power, communication and signal cables running external to the building. Surge protection devices shall be installed at both ends of the line, as close to the device being protected as possible and in the same enclosure.

(d) Surge protection devices shall be in accordance with IEC 61643-11.

(e) Surge protection devices shall have plug-in protection modules.

(f) Surge protection devices shall contain multiple means of transient protection including:

(i) Gas tube for high current diversion;

(ii) Zener diodes for voltage clamping;

(iii) Series resistive elements for current limiting; and for power-line devices; and

(iv) Metal oxide varistors (MOV).

(g) Transient rating of surge protection devices shall be 10 kA (8/20 µs waveform) with a response time of less than 10 ns, and devices shall be self-resetting and shall have external indication of condition.

(h) Surge protection devices shall be as follows:

(i) Power Supply Line Protection

1) For power supply line protection, the surge suppression units shall provide full protection against all possible transient modes (phase to neutral, phase to earth, neutral to earth). The transient let through voltage shall not exceed 720 V with a surge current of 3 kA. The protection devices shall be rated for a peak discharge current of 20 kA with a phase to earth leakage current of less than 60 uA. The protection devices shall be provided with facilities for continuous status indication for “full protection present”, “reduced level of protection”, “protection failure.” Power supply line surge suppression units shall be of the plug-in type which has a fixed base and terminals with a plug-in suppression unit. Power-line surge protection devices shall be connected via HRC fuses to facilitate isolation for maintenance. HRC fuses shall comply with IEC 60269.

(ii) Signal and Data Line Protection

1) For data and signal line applications the maximum let-through voltage tested under a 6 kV/ 3 kA (8/20 µs waveform) combination waveform shall be less than 100 V and when tested with a 1 kV/µs waveform the maximum let through voltage shall be 40 V. Dual channel surge arrestors shall be used to conserve space in cabinets. Surge arrestors shall feature a means of visually identifying units that need replacing.

(i) Surge protection devices shall include a mechanism for the remote monitoring via a single volt free contact, of up to at least 20 protection devices mounted on the same DIN mounting rail.

G6.2.4 Instrument Earth

(a) A system of separate instrument earth conductors shall be provided for ICA equipment. This shall be known as instrument earth (I.E.). The instrument earth shall also be used by any other equipment requiring a “clean” earth.

(b) The instrument earth shall have its own electrodes and shall be connected to the electrical earth system at the main earth bar within each facility via a surge protection device.

(c) Instrument earthing shall comply with SS 551.

(d) The physical earthing arrangements and materials shall be as per the Electrical section of the General Specification.

(e) Labels shall be provided for each instrument earth bar to differentiate it from electrical earth.

G6.2.5 Uninterruptible Power Supplies

(a) UPS units shall comply with the requirements of General Specification G5 Electrical Works and Particular Specification P6 ICA.

(b) The UPS DB shall be arranged such that single points of failure for the plant control system are avoided. No residual current device (RCD) shall control the power to more than one server or one operator workstation.

G6.2.6 Miscellaneous

(a) Equipment shall not be mounted on handrails or in the access way where this results in an increased risk of physical damage to the equipment.

(b) All equipment shall be provided with proper access for maintenance.

G6.2.7 Instrumentation Installation

(a) All instruments shall be installed in accordance with industry best practice and the manufacturers recommendations, and in accordance with standard installation details included in the Drawings.

(b) Precise instrument locations shall be shown on the Contractors final submittals including within the BIM model updated by the Contractor.

(c) Accessibility and maintainability shall be given careful consideration and instruments and actuators requiring adjustment shall be accessible from floor level or have suitable access arrangements provided.

(d) The Contractor shall provide all miscellaneous work associated with the correct and appropriate installation of instruments. For example, stilling wells.

(e) Analysers installed outdoors shall be housed within outdoor panels built in accordance with General Specification G6.4 Instrument and Control Panel Construction.

(f) Any drains from instrument racks or analysers shall be routed to the nearest plant drain and shall not discharge to floor.

(g) The Contractor shall use the instrument manufacturers' mounting brackets and installation hardware in all cases unless these are inferior to the Specification with regard to material selection for corrosion resistance.

G6.2.8 DC Uninterruptible Power Supply (UPS) Units

(a) DC UPS units may be mounted within an ICA panel and shall consist of a UPS control unit and a battery.

(b) Each DC-UPS control unit shall be DIN-rail mounted and shall have the following features:

(i) Input Voltage : 24 Vdc;

(ii) Output Voltage: 24 Vdc;

(iii) Output Current: 20A;

(iv) Operating Temperature Range: -40deg C to +70 deg C;

(v) Efficiency: 99%;

(vi) To suit nominal battery voltage: 2 x 12 V (series connected with centre tap);

(vii) To suit nominal battery capacity: up to 120 Ahr (24 V);

(viii) Multi-mode charger;

(ix) Temperature compensated with battery temperature sensor;

(x) Electronically protected against output overloads;

(xi) Automatic battery health diagnostics; and

(xii) Relay contacts for monitoring.

(c) Batteries for DC-UPS units shall be as follows:

(i) Voltage: 2 x 12 V (series connected with centre tap);

(ii) Capacity: To suit load, up to 120 Ahr; and

(iii) Chemistry & Construction: AGM Lead-Acid.

END OF SECTION